FUSION ERP LAB ASSIGNMENT 6B

Phase 2 — Review, Test and Improve

Employee Leave Management (ELM)

Repository: Fusion-Integrated

Phase-2 branch: `elm-phase2`

Phase-2 commit: `c3ab25c`

Preserved Phase-1 baseline: `d1cb96a`

Upstream pull request: FusionIIIT/Fusion-Integrated#359

Verification date: 27 October 2026

Conclusion: Phase 2 delivered and verified meaningful improvements to employee self-service, request lifecycle integrity, and overlap validation. ELM is not yet deployment-ready against the complete specification; substantial business workflows and external contracts remain unresolved.

Prepared from the Phase-2 implementation, automated test results, local runtime checks, the supplied ELM specifications, and inspection of the Administration package.

1. Objective and approach

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Assignment 6B asks the team to treat a Phase-1 AI-generated ELM module as inherited software: understand it, test and review it against the complete ELM specifications, improve it while preserving its original version, and verify the result.

The inherited ELM implementation was a small backend leave-request API without an employee-facing ELM workflow. Phase 2 continued from the preserved Phase-1 baseline in a separate branch. The specifications were reviewed for lifecycle, overlap, half-day, printable-application, entitlement, calendar, hierarchy, substitute, and post-approval workflow requirements.

Specification sources reviewed

- ELM Problem Statement v1.4
- ELM Use Case Specifications v1.4
- ELM Business Rule Specifications v1.4
- ELM Functional Requirement Specification v1.5 and traceability materials
- ELM Workflow Specifications v1.6
- Fusion Base Code contracts and the supplied Fusion System Administrator package

Fusion integration surfaces changed

- ELM server registry and role grants for employee leave-request capabilities.
- Server-driven navigation and client route registration for the ELM page.
- API serializers, viewset behavior, leave data model, event history, and migrations.
- OpenAPI schema, generated permission catalogue, and repository README.

2. Improvements delivered

Area	Change	Specification value
Employee self-service	Employees can submit and view their own leave requests through the Fusion client.	Establishes an integrated user workflow rather than API-only functionality.
Request ownership	Queries are scoped to the authenticated employee; other users' requests are not exposed.	Protects employee request privacy and ownership.
Lifecycle integrity	Submitted requests cannot be updated or deleted. Employees can immediately withdraw only their own pending request.	Implements withdraw-and-resubmit behavior and prevents withdrawal after a decision.
Audit history	Submission and withdrawal transitions are persisted as events and returned with requests.	Provides an inspectable history for implemented state changes.
Overlap validation	Rejects date overlap with pending or approved leave. Rejected/withdrawn requests do not block new requests.	Implements active-leave non-overlap for whole-day requests.
Half-day CL	Supports full day, first half, or second half for a single-day Casual Leave; opposite halves can coexist.	Implements the specified same-date different-half CL exception, validated in API and database.

Area	Change	Specification value
Printable application	Print view includes request number, employee ID, category, dates, reason, current status, submission timestamp, and event history.	Provides a printable representation for the implemented application data.
Contracts and verification	Updates permissions, navigation, migrations, OpenAPI, documentation, and regression tests.	Keeps the Fusion-facing contracts aligned with the implementation.

3. Verification results

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Check	Result	Notes
Full backend test suite `.venv\Scripts\python.exe -m pytest -q`	**PASS**	Complete repository pytest suite passed.
Client typecheck `cd client; npm run typecheck`	**PASS**	No TypeScript diagnostics.
Client tests `cd client; npm test -- --run`	**PASS**	4 files, 35 tests passed.
Production build `cd client; npm run build`	**PASS**	Vite build succeeded; existing >500 kB chunk warning remains.
Django configuration `manage.py check`	**PASS**	No system-check issues.
Migration consistency `manage.py makemigrations --check --dry-run`	**PASS**	No uncommitted model migrations detected.
Database migration `manage.py migrate --noinput`	**PASS**	ELM migrations 0004 and 0005 applied to the configured local database.
ELM lint `ruff check modules/elm`	**PASS**	No Ruff findings.
OpenAPI generation	**PASS**	Schema generated with warnings treated as failures.
Local frontend	**PASS**	Vite responded HTTP 200.
Local protected API	**EXPECTED 401**	Unauthenticated ELM API request was rejected as expected.

ELM-specific regression scenarios

- Staff and faculty self-submission; submitted owner is taken from authenticated identity.
- Requests belonging to another employee are not readable, withdrawn, or deleted by the caller.
- Permission and module grants are enforced.
- End date preceding start date is refused.
- Overlaps with both pending and approved leave are refused; rejected leave does not block.
- First-half and second-half single-day CL are accepted together; duplicate half and other overlaps are refused.
- Half-day selection is refused for non-CL or multi-day requests.
- Submitted requests are immutable; pending requests can be withdrawn and history records both transitions.

4. Runtime and integration limitations

> **Authentication integration was not end-to-end verified.** The local app is configured to contact the ERP-user IAM at `http://127.0.0.1:8001`. No service was listening on that port during the local run. The supplied Administration package documents a separate operator console and does not provide the ERP-user IAM login contract required by this application.

The backend started and passed Django checks. The frontend served successfully. An unauthenticated call to the protected ELM endpoint returned 401, which confirms the authorization boundary is active, but does not demonstrate a successful logged-in user journey. The client lint script was not verified in this run because the project's ESLint executable was unavailable in the earlier check.

5. Important unresolved specification gaps

Gap	Impact / dependency
Leave entitlements, balances, account history, lapse/carry-forward, and year-end conversion	Core financial/accounting-like leave correctness remains absent; requires policy rules and a ledger.
Holiday/RH and Academic Calendar vacation validation	Cannot safely validate RH dates or VL periods without authoritative published calendar services/data.
Unit Head and higher-sanction approval hierarchy	Department/designation data alone does not establish current Unit Head or route. No authoritative hierarchy contract was found; routing must not be guessed.
Substitute nomination, availability, consent, decline, and replacement	Required coverage gate before approval routing is absent.
Approver recommendations and final decisions	No actionable Unit Head / Establishment / Sanctioning Authority workflow exists.
Cancellation, extension, and resumption	Approved leave changes, balance adjustments, and attendance synchronization are absent.
Administrative maintenance, offline-sanctioned entries, SLA, and notifications	Configuration, retrospective integrity, pending-task escalation, and communications remain absent.

6. Overall assessment

> **Phase-2 progress: verified partial improvement.** The inherited implementation now has a functional, authorization-scoped employee request slice, stronger lifecycle and overlap rules, a printable representation, and regression evidence. Phase 1 remains preserved at `d1cb96a`; Phase 2 is at `c3ab25c`.

Deployment readiness: not achieved. Major leave rules, approval workflows, calendar/hierarchy integrations, and authenticated runtime validation remain open. No claim is made that the module satisfies the complete ELM specification or is production-ready.

Detailed supporting files: `verification-evidence.md` and `issue-improvement-log.md` in `docs/elm-phase2/`.